

Invisible Video Watermark Method Based on Maximum Voting and Probabilistic Superposition

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ABSTRACT

Invisible watermarking is an essential measure for media publishers to declare ownership of their content, in the case of minimizing the impact on the viewing experience. In dealing with active attacks such as noise attacks, filtering attacks, geometric attacks, and lossy compression attacks, existing research still has great limitations. In this paper, through probabilistically superposition of "perturbed watermark obtain by maximum voting method" and "determined Bernoulli distribution" to restore the real embedded watermark, is used to deal with complex attack situations. Specifically, the special embedded watermark is obtained by sampling from the n -fold Bernoulli distribution with parameter p . Secondly, the *HAAR* wavelet transform is performed on the *YUV* channel of the video fixed-interval image to extract its low-pass component. Then Discrete Cosine Transform is performed to obtain its frequency domain representation. The watermark information is embedded into the frequency domain representation's singular to realize the embedded invisible watermark of video. The watermark bit of every block is determined by the maximum voting method for the disturbed watermark that performs *DCT* and *SVD* operations on the low-pass component of *YUV* channels. At this time, the determined Bernoulli distribution is probabilistically superimposed on the watermark information to guarantee distribution consistency. Finally, mean value operation and cluster processing are performed on the watermark information to reduce volatility. Experiments show that the method proposed in this paper has apparent advantages in solving Invisible video watermarking. With a PSNR of 41 and a corresponding BAR of 0.86, our team achieves 3rd place on the leaderboard of the Invisible Video Watermark Challenge 2023.

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CCS CONCEPTS

• Mathematics of computing → Probability and statistics.

KEYWORDS

Invisible video watermark, Maximum voting, Probabilistic superposition

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1 INTRODUCTION

In recent years, digital watermarking technology has been widely used to protect the copyright of digital products, to prove the authenticity of products, to track piracy, or to provide additional product information [8, 19]. However, many characteristics of digital watermarking algorithms are mutually restricted, such as imperceptibility, space efficiency, robustness, security, reversibility, time efficiency, and space efficiency [17]. For example, the well-known spatial domain algorithm, Least Significant Bit(LSB), has good embedding capacity and low computational complexity. Due to the principle of embedding, this method is less robust and lacks resistance to common attacks. The transform domain watermarking algorithm makes up for this shortcoming to a certain extent, transforming the image into a different domain, and then embedding the watermark by modifying the coefficients, this method improves the resistance of the watermark to standard attacks and the robustness of the watermark [6, 18].

With the development of technology, the emergence of various attack types puts forward higher requirements for the digital watermark embedding algorithm [3]. In dealing with active attacks such as noise attacks, filtering attacks, geometric attacks, and lossy compression attacks, existing research still has excellent limitations. Solachidis et al.[16] used the scaling and rotation invariance of the Fourier domain to propose an embedding watermarks algorithm, that has good robustness against filtering attacks and

compression attacks. Qi Xiangming et al.[15] proposed a watermarking algorithm based on *DWT* and *SVD*, which achieves a better ability to resist geometric attacks, but the ability to resist shear attacks is average. Xu Jiangfeng et al.[7] proposed a watermarking scheme that integrates chaotic systems, *DWT*, and *DCT* to improve transparency, but its resistance to shearing attacks was found to be inadequate. Guo Pengfei et al.[4] proposed a digital watermarking algorithm based on redundant discrete wavelet transform combined with Schur decomposition which exhibits limited robustness against rotation attacks and inadequate resistance to shearing attacks. Mohammed et al.[13] proposed a semi-blind image watermarking scheme based on the *DWT-SVD* transform domain, the watermark position is embedded in a "Z" shape, which improves the imperceptibility. This scheme achieves good transparency and robustness, but its robustness against attacks such as lossy compression and median filtering is average.

In this paper, we restore the real embedded watermark perturbed by the probabilistic superposition of the "embedded watermark obtained by maximum voting method" and "determined Bernoulli distribution." The unique embedded watermark is obtained by sampling from the n -fold Bernoulli distribution with parameter p . Secondly, the *HAAR* wavelet transform is performed on the *YUV* channel of the video fixed-interval image to extract its low-pass component, the discrete cosine transform is performed on the fixed block of the low-pass part to obtain its frequency domain representation, and the watermark information is embedded into the singular value of the frequency domain representation, and then perform a series of inverse operations to realize the embedded video invisible watermark. Then perform watermark extraction on the fixed-interval image of the disturbed video. Perform *DCT* and *SVD* operations on the fixed block of the low-pass component of the *YUV* channel to obtain the concerned watermark information. The watermark information of every block is determined by the maximum voting method for the watermark information of the three channels of *YUV*. At this time, the determined Bernoulli distribution is probabilistically superimposed on the watermark information to guarantee distribution consistency. Finally, mean value operation and cluster processing are performed on the watermark information.

2 RELATED WORK

The transform domain watermarking algorithm makes good use of the signal characteristics and the characteristics of human perception [2]. The digital watermarking algorithm based on this method has better imperceptibility and stronger robustness [10]. Generally speaking, the frequency domain is more suitable for digital watermarking algorithms. Common transform domain watermarking algorithms include Discrete Fourier Transform, Discrete Cosine Transform, and Integer Wavelet Transform.

2.1 Discrete Fourier Transform

DFT realizes the mapping relationship between the spectral function and the signal, where the independent variable of the signal is time, and the independent variable of the spectral function is frequency. In the *DFT* process, the independent variables time and frequency are discrete [9]. The specific formula of *DFT* and its inverse transformation is as follows:

$$X(K) = \text{DFT}(x(n)) = \sum_{n=0}^{N-1} x(n)e^{j(2\pi/N)nK},$$

$$x(n) = \text{IDFT}(X(K)) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)e^{-j(2\pi/N)nK},$$

Where $X(K)$ and $x(n)$ represent the primary value sequence of the frequency domain and the principal value sequence of the time domain signal, respectively. $\text{DFT}(\cdot)$ and $\text{IDFT}(\cdot)$ stand for forward and inverse discrete Fourier transform respectively. K represents the number of sequences in different frequency bands, N represents the length of the line.

2.2 Discrete Cosine Transform

Discrete cosine transform can compress and store a large amount of information in low-frequency components, while other components, such as high-frequency components, store very little information [1, 11]. The *DCT* transformation formula for a two-dimensional image $f(i, j)$ of size $N * N$ is as follows:

$$F(u, v) = c(u)c(v) \frac{2}{N} \times \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} f(i, j) \cos\left(\frac{2i+1}{2N}u\pi\right) \cos\left(\frac{2j+1}{2N}v\pi\right),$$

Where $F(u, v)$ is the frequency domain information of image $f(i, j)$ after *DCT* transformation. For the inverse transform of *DCT*, the formula is as follows:

$$f(i, j) = \frac{2}{N} \times \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} c(u)c(v)F(u, v) \cos\left(\frac{2i+1}{2N}u\pi\right) \cos\left(\frac{2j+1}{2N}v\pi\right),$$

$$c(u) = c(v) = \begin{cases} \frac{1}{\sqrt{2}}, & u = 0, v = 0 \\ 0, & \text{otherwise} \end{cases}$$

2.3 Integer Wavelet Transform

The *IWT* can be considered a special realization of the discrete wavelet transform. The calculation complexity of general discrete wavelet transform is high, and the calculation and rounding of floating point numbers will bring calculation errors [14]. However, *IWT* inherits the multi-resolution characteristics of the traditional wavelet transform and can map the input data to integers without quantization errors. In addition, the *IWT* transform requires less storage space, and the implementation of the algorithm is faster than the traditional wavelet transform [12]. *IWT* consists of three parts: separation, prediction, and update.

Separation: Decompose the input sequence S_j to get even sequence $Even_{j-1}$ and odd sequence Odd_{j-1} . Among them, $Even_{j-1} = \{Even_{j-1,k} = S_{j,2k}\}$, $Odd_{j-1} = \{Odd_{j-1,k} = S_{j,2k+1}\}$.

Prediction: One of the sequences Odd_{j-1} is based on another sequence $Even_{j-1}$ predicted according to the correlation of the data. The difference D_{j-1} between the actual value Odd_{j-1} and the predicted value $P(Even_{j-1})$ as the next layer's odd numbers sequence of *IWT*.

Update: After the prediction comes the update step. Combining the updated odd samples with the even samples results in updated even samples. $S_{j-1} = Even_{j-1} + U(D_{j-1})$, where U is the update operation, the formula is as follows:

$$U_k(D_{j-1}) = \frac{D_{j-1,k}}{2} = \frac{D_{j-1,k-1} + D_{j-1,k}}{4} + \frac{1}{2}$$

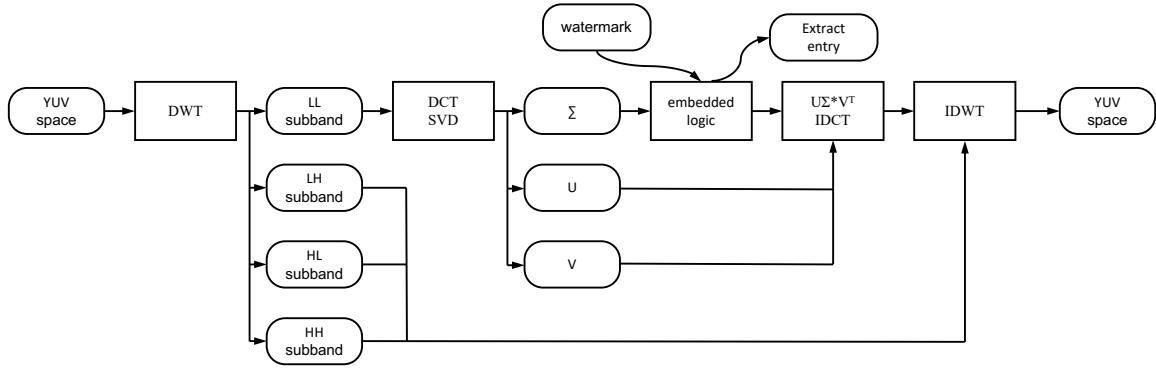


Figure 1: Overview of watermark embedding and extraction process. Watermark embedding of the transform domain is realized through *DWT*, *DCT*, and *SVD* algorithms. Every bit of watermark information is cyclically embedded into the fixed block of the *LL* subband at the *YUV* channels. The bit watermark information of every block is determined by the maximum voting method for the disturbed watermark on the low-pass component of *YUV* channels. The determined Bernoulli distribution is probabilistically superimposed on the watermark information, then mean value operation and cluster processing are performed to realize extraction.

3 PROPOSED METHOD

As shown in Fig. 1. In this paper, based on the watermark algorithm of *DWT*, *DCT*, and *SVD*, the determined Bernoulli distribution probability is superimposed on the disturbed watermark information to resist complex active attacks. The method proposed consists of three parts: watermark generation, embedding, and extraction.

3.1 Watermark Generation

If the random variables in the random variable sequence X_n , ($n = 1, 2, \dots$) all obey the Bernoulli distribution with parameter p , then the random variable sequence X_n forms an n -fold Bernoulli experiment with parameter p .

The generation of watermark W is defined as an n -fold Bernoulli sampling process with parameter p :

$$Wm = [wm_1, wm_2, wm_3, \dots, wm_{256}]$$

Obviously, the watermark is a fixed-length random 01 sequence, and the parameter p limits the randomness of the watermark. To ensure the randomness of the watermark sequence while considering its singularity, the parameter p is not a fixed value but is limited within a specific range. This Bernoulli distribution with parameter p is called a "determined Bernoulli distribution."

3.2 Watermark Embedding

To balance video quality and embedding efficiency, we only perform watermark embedding on images of fixed intervals in the video. The first step is to perform color space conversion on the sampled image, convert it from *RGB* format to *YUV* format: $[Y, U, V]^{-1} = M \cdot [R, G, B]^{-1}$, where M is a fixed parameter matrix, Y represents the luma component, and U and V represent the chroma components. Then *HAAR* wavelet transform is performed to decompose the *YUV* channel data into sub-bands of different frequencies and obtain each channel's low-frequency and high-frequency components at different scales. In detail, the data of each channel is

segmented into matrices composed of rows and columns. A one-dimensional wavelet transform is performed on each row to obtain the low-frequency and high-frequency components in the horizontal direction. Subsequently, a one-dimensional wavelet transform is applied to each column of the transformed data. The wavelet transformation formula is as follows:

$$X_f(a, b) = \int_{-\infty}^{+\infty} f(t) \cdot \frac{1}{\sqrt{a}} \psi \left(\frac{t-b}{a} \right) dt$$

After *DWT* transformation, each channel is divided into four non-overlapping subbands, namely: *LL*, *HL*, *LH*, and *HH*. These subbands contain image information in different frequency ranges. Among them, the low-frequency sub-band *LL* preserves most of the information content of the original image, so it has stronger anti-interference and stability and is less susceptible to noise and distortion.

Based on this, we divide the *LL* subband of each channel into blocks according to the size of 4×4 and perform a discrete cosine transform on each block to obtain the *DCT* coefficient matrix F .

$$F(u, v) = c(u)c(v) \frac{2}{N} \times \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} f(i, j) \cos \left(\frac{2i+1}{2N} u\pi \right) \cos \left(\frac{2j+1}{2N} v\pi \right),$$

Where $F(u, v)$ represents the element of row u and column v of *DCT* coefficient matrix F .

Perform singular value decomposition on the coefficient matrix $F = U \cdot S \cdot V^T$, Embed the watermark according to $S_0 = (S_0/d_1 + 1/4 + 1/2 * Wm_i) * d_1$, where d_1 is the strength of the embedded watermark. Then perform a series of inverse operations including *ISVD*, *IDCT*, and *IDWT* to realize the embedding of watermark information. It is important to note that the complete watermark consists of 256 bits, but only 1 bit of watermark information is embedded in each 4×4 block. Therefore, the watermark information is cyclically embedded into the *LL* subband of the *YUV* channels.

3.3 Watermark Extraction.

Videos embedded with watermarks may be vulnerable to various random active attacks, such as resolution, rotation, noise, and crop attacks. These attacks can potentially degrade the quality or alter the appearance of the watermarked videos. The steps to get to the extraction entry are the same as the embedding process: Convert the image from *BGR* to *YUV* format, perform *HAAR* wavelet transform on the *YUV* channel, perform 4×4 block processing on the low-pass component *LL* after wavelet transform, and perform discrete cosine convert and singular value decomposition on each embedded block.

In the case where a watermarked video has been subjected to random attacks, performing watermark extraction operations ($Wm_i^Y = (S_0 \% d_1 > d_1/2) * 1$) can only yield perturbed watermark information on the blocks. We perform watermark extraction on the three channels of *YUV* at the same time, and the watermark information of each block is obtained by the maximum voting method. The watermark information on the *i*-th block is $Wm_i' = \text{Voting}(W_i^Y, W_i^U, W_i^V)$.

Considering many active attacks, the extracted watermark information will be damaged, especially the distribution. Therefore, we try to fix this perturbed distribution using the Bernoulli distribution of sampled watermarks. However, if the Bernoulli distribution of the sampled watermark is too random and too common, the repair effect will be poor. Therefore, when sampling the watermark, its parameter *p* is limited within a specific range, see section 3.1 for details. Next, we superimpose a specific Bernoulli distribution on the watermark information of each block or every 256 blocks (complete watermark) with the probability β .

$$Wm_{i:j}^p = \begin{cases} Wm_{i:j}' & P \leq \beta \\ (Wm_{i:j}' + Wm_B)/2 & P > \beta \end{cases}$$

Where *P* is the probability of random, Wm_B is sampled from a definite Bernoulli distribution of the watermark generation process. Here Wm_i' guarantees the authenticity of the watermark information, even though it is perturbed. The determined Bernoulli distribution ensures the consistency of the watermark distribution as much as possible.

Since the fixed-length watermark information is cyclically embedded into the corresponding blocks, the length of the extracted watermark sequence is much larger than this length. Therefore, it is necessary to perform a mean value operation on the extracted watermark sequence and restore it to a fixed length. Finally, clustering is performed to make the watermark information 0 or 1.

4 EXPERIMENTS

4.1 Dataset

The copyright of Mango TV protects all of the data provided in this challenge. The Organizer divides the dataset into two parts: the public and testing datasets.

The public dataset consists of 900 video clips with a resolution of 720×1280 or 480×848 , each lasting about 30 seconds, and the testing dataset of 100 video clips with the same characteristics, which is used for evaluation and not publicly released.

Table 1: Ablation studies of our method on the Test Dataset.

Baseline Model	Maximum Voting	Probabilistic superposition	PSNR	BAR
✓			41.17	0.519
✓	✓		40.72	0.629
✓		✓	41.49	0.708
✓	✓	✓	41.33	0.862

4.2 Metrics

Evaluation metrics include extraction accuracy and watermarked video quality.

Watermarked video quality: The peak signal-to-noise ratio (PSNR) is to be considered when measuring the quality of watermarked videos. A video's PSNR score is calculated as the average of all its frames' PSNR, and the final PSNR is calculated as the average of all videos. The PSNR for a single frame is computed as follows:

$$PSNR = 10 \log_{10} \frac{MAX_i^2}{MSE}$$

Extraction accuracy : This challenge adopts 256-bit as the watermarks for each frame, and the bit accuracy ratio (BAR) is used to measure the accuracy of extracted watermarks.

$$BAR = \frac{1}{X} \sum_{1 \leq i \leq X} D(WM_O, WM_E(i))$$

4.3 Performance

As shown in Table 1. With a PSNR of 41 and a corresponding BAR of 86%, our team achieved outstanding results in the Invisible Video Watermark Challenge 2023, placing among the top performers on the leaderboard. Ablation experiments demonstrate the effectiveness of the "Maximum Voting" and "probabilistic superposition" components.

5 CONCLUSION

This paper uses probabilistically superposition of "perturbed watermark obtained by maximum voting method" and "deterministic Bernoulli distribution" to restore the real embedded watermark to deal with complex attack situations. It is worth pointing out that the extracted watermark guarantees the certainty and authenticity of the watermark information, even though it is perturbed. The determined Bernoulli distribution ensures the consistency of the watermark distribution as much as possible. It is essential to balance the "perturbed watermark" and "deterministic Bernoulli distribution." However, this method has a certain degree of limitation, which limits the randomness of the watermark. We will continue improving our solution in the future, looking for a better watermark generation scheme.

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